

QINGLIU WU



☎ 17353675760 · ✉ wql@bupt.edu.cn · 🌐 micuks

EDUCATION

Beijing University of Posts and Telecommunications (BUPT) 09/2024 – 06/2027

- M.Sc. Candidate in Computer Science; **National Scholarship (2024–2025)**; AI4DB research at the IDMG Lab under Prof. Yingxia Shao.

BUPT 09/2020 – 06/2024

- B.Eng. in Computer Science (Experimental Class); score 91.91, ranked 29/422 (top 6.56%).

INTERNSHIP EXPERIENCE

Key Multimodal Model Pre-training Infrastructure / *Megatron* 2026.05 – Present
Kuaishou, Foundation Model & Application Dept. / AI Infra

- Implemented Context Parallel for Causal-ViT long-video pre-training; integrated Ulysses All-to-All for GDN linear attention and cross-shard halos for MoDA; scaled CP8 single-sequence training from 131K to 2M tokens on H-class GPUs and designed a CP16 plan for 4M tokens.
- Replaced a launch-bound per-frame Python loop (~80% GPU idle) with cross-frame batching; matched forward/backward results bitwise and achieved $5.4\times$ layer / $1.28\times$ end-to-end speedups. Tiled halo reprojection decoupled memory from sequence length; dependency-planned direct P2P replaced fixed one-hop r_{i-1} halos, supporting short/empty shards without the $T \geq W \times sp_size$ constraint.
- Removed the CP batch-size-1 restriction caused by single-sample GDN Ulysses calls and sample-local MoDA halo/loss metadata. Designed a padding-free **PackedVisionLayout** with per-sample **cu_seqlens** and frame ownership to drive GDN All-to-All, MoDA halos, and loss reductions; enabled mixed image-video batches and packing with BF16 numerics, loss, and grad-norm curves aligned.
- Root-caused Muon incompatibility with ZeRO-1/2 reduce-scatter: matrix-level Newton-Schulz updates lacked complete reduced gradients for partitioned matrices. Upstreamed owner-directed sliced propagation and redundant owner updates to DeepSpeed; **PR#8141 merged into master**.
- Ensured DeepSeek-V4-Flash pre-training correctness: isolated fused RoPE as the source of BF16 NaNs and filed Megatron-LM#5317 with a minimal repro; traced a 10^{-3} weight-conversion drift to FP32 parameters being downcast to BF16 and restored tensor-wise parity.

Flink Large-State Processing Optimization 2026.03 – 2026.05
Huawei, Computing Product Line / AI Engineer

- Built two-tier caches for Flink 1.16.3 ValueState/MapState and backpressure-aware async prefetch with RocksDB MultiGet, reducing backend accesses and synchronous wait.
- Pre-aggregated small state-operation batches, coalescing multiple read-modify-write requests to reduce JNI overhead.
- Achieved 30% higher average throughput than RocksDB StateBackend on a 100M-scale Nexmark large-state subset and passed CDC consistency validation.

PROJECTS

SQLForge: Text-to-SQL Agent Post-training / *Execution Feedback / QLoRA / GRPO* 2026.01 – 2026.05

- Built an Agent post-training pipeline with multi-teacher ReAct trajectory distillation, real SQLite observations and recovery traces, QLoRA SFT, and a custom GRPO loop with dense format/validity/structure/execution rewards.
- Enabled iterative SQL execution and self-correction in a SQLite sandbox; improved Spider Execution Match from 66.9% to 78.6% (+11.7pp), with +13.4pp on Hard queries.
- Teacher-forced top-k analysis found the gold SQL within the top-50 at every step for 98% of cases; K=8 execution-result voting beat a single sampled rollout by 6.5pp (95% CI: +4.6 to +8.4), identifying decoding selection as the remaining bottleneck.

vLLM Online Inference Scheduling Optimization / *Python / vLLM / GPU Serving* 2026.01 – 2026.02

- Replaced FCFS with an adaptive SJF-priority MLFQ plus low-load bypass to mitigate head-of-line blocking: on Qwen3-4B at RPS 128, improved goodput by 35.5% and throughput by 21.0%, while reducing p95 TTFT by 44.8% with no low-load regression.

RESEARCH EXPERIENCE

DBPecker: Compound Anomaly Diagnosis for Distributed Databases PVLDB 2025 Demo · Co-first Author

- Developed graph-propagation-based compound anomaly localization and structural-causal root-cause ranking across layers and nodes; validated end-to-end through fault injection on an OceanBase cluster and integrated LLM-generated troubleshooting guidance.

SKILLS

- **LLM Algorithms:** SFT, GRPO reinforcement learning, ReAct Agent design, training, and evaluation.
- **Training & Serving:** PyTorch, Megatron-LM, DeepSpeed, vLLM; pre-training debugging and inference optimization.
- **Languages & Tools:** C++/Python; familiar with Java/Go/Rust; Linux, Git, Docker, CMake.
- **English:** IELTS 7.5 (Reading 8.5, Listening 8.5); CET-6 594.
- **Other:** National First Prize, 24th China Robotics and Artificial Intelligence Competition - Baidu Apollo Autonomous Driving Simulation (2022).